

ROOTMESH · REQUIREMENTS

RootMesh requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/rootmesh		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-02	Amish Chadha	Follow-ups of 2026-10-02: R5 margin on US915, R9 cell chosen, R12 cost with the status LED; no status changed
0.6	2026-10-02	Amish Chadha	R9 cell limit restated to the chosen cell's rated discharge range (55 °C or more), charging blocked above 45 °C, as decided by Amish on 2026-10-02
0.5	2026-10-01	Amish Chadha	Figures for the constructable design (RMS-DDR-003, RMS-CAL-001 v0.3); budget treated as a value-engineering target; no status changed
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	Apply Amish's decisions (RMS-DDR-001); redefine R4, R5, R11, R12 and R13; status from RMS-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status for each
0.1	2026-09-24	Amish Chadha	Initial scaffold

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These requirements reflect Amish's decisions of 2026-09-25 (RMS-DDR-001 and RMS-DDR-002) and the constructable design of RMS-DDR-003 and are checked by calculation in RMS-CAL-001. Targets are not yet validated with growers and will be revised after user input (see RMS-PRB-001; first users are still open). Nothing has been built or measured, so "met" means met on paper, by calculation or by a datasheet.

Reference case: one stake in a loam field, sensing at 150 and 300 mm, reporting every 20 min at SF10 to an indoor gateway about 1 km away, with the antenna at about 1 m on the marker rod, in a region with 3 peak sun hours per day in the worst month.

Table 1. Requirements and TRL 3 status (RMS-CAL-001, Table 6).

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3
R1	Measure volumetric water content at two root-zone depths	150 mm and 300 mm below grade (± 25 mm); 5 to 50 % VWC; ± 3 % VWC after a site-specific calibration	Model geometry; later gravimetric calibration in two soils	At risk: depths met by the model; accuracy depends on probe contact and calibration
R2	Measure root-zone soil temperature	± 0.5 °C from -10 to 60 °C at about 225 mm depth	DS18B20 datasheet; model geometry	Met (datasheet; 225 mm in the model)
R3	Measure bulk soil electrical conductivity	0 to 4 dS/m; ± 10 % or ± 0.1 dS/m, whichever is larger, after calibration in standard solutions; reported at 25 °C	Circuit calculation; later bench test in KCl solutions and soil	Met on paper: circuit error under 2 % (RMS-CAL-001, D)
R4	Report often enough for daily irrigation decisions	Every 20 min by default (RMS-DDR-001, D9); adjustable from 10 to 60 min where the spreading factor keeps uplink airtime at 30 s or less per day; the default fits at SF10	Airtime calculation	Met: 26.7 s per day at SF10 and 20 min; 15 min needs SF9 or faster
R5	Reach the gateway	90 % or more of uplinks received at 1 km from an indoor gateway, with the stake antenna at about 1 m on the marker rod (RMS-DDR-001, D2), open field with a mature crop	Link budget; later field range walk	At risk: 19.5 dB margin at 1 km on US915 (20 dBm radiated), but a crop taller than the antenna can take 13 to 19 dB, which leaves 0.1 dB at 100 m of crop
R6	Run on sun alone	Energy-neutral with 1 peak sun hour per day on the panel; 90 days or more of reporting with no harvest at all	Energy budget	Met: 19 times daily need at 1 sun hour; 186 days on the cell alone
R7	Safe cell and charging	LiFePO4 chemistry, 2 Wh or less per stake; charging blocked below 0 °C and above 45 °C cell temperature; cell fused	Design review; charger datasheet	Met by design (1.92 Wh, NTC cut-off, PTC fuse)
R8	Keep water out	Head IP67; buried section IP68 at 0.5 m for a growing season; UV-stable exposed parts	Design review; later immersion and spray test	Not verifiable at TRL 3; vent membrane added against 15 kPa daily thermal pumping
R9	Operate across field temperatures	-10 to 60 °C at the head; at the cell, the chosen cell's rated discharge range, which its datasheet must rate to 55 °C or more, with charging blocked above 45 °C (restated from -5 to 40 °C by Amish, 2026-10-02; if the cell is not rated to 55 °C, a printed shade skirt round the head is added)	Component datasheets; thermal estimate	At risk: head about 64 °C and cell up to about 47 °C in bare, hot soil (parts rated 85 °C); inside the chosen IFR14500EC class cell's datasheet discharge range of -20 to +60 °C (RMS-DEC-001; no shade skirt needed); the head stays over its 60 °C
R10	Install and remove by hand	One person with a 50 mm hand auger and the slot tool (RMS-DDR-002) installs a stake in 15 min or less, with sensing	Push-force estimate; later timed trial	At risk: with the slot tool about 386 N in moist loam (within one person's 500 N), about 1.2 kN in firm dry

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3
		depths repeatable to ± 25 mm; removal by hand in 5 min or less		loam (without the tool 526 N and 1.8 kN); auger hole 110 mm deep (RMS-DDR-003)
R11	Survive field operations	Rigid head 300 mm or less above grade; anything higher, including the antenna (RMS-DDR-001, D2), carried on the flexible marker rod; high-visibility marker 1 m or more above grade	Model check	Met: head 171 mm; rod top 1,000 mm; antenna 914 to 1,086 mm on the rod
R12	Affordable	One stake \$60 or less in parts; pilot set of three stakes, one gateway and the installation slot tool \$300 or less, which is how the \$300 value-engineering target is read (RMS-DDR-001, D4)	Priced BOM (bom/ bom.csv)	Met: \$59.15 per stake; \$281.45 per pilot set including the slot tool, USD 18.55 under the value-engineering target (indicative, RMS-DDR-003 and the status LED of 2026-10-02)
R13	Grower owns the data	Data readable without a paid subscription; CSV export; open dashboard. The pilot uses The Things Network community server (RMS-DDR-001, D3)	Design review	Met by design; a fully local server with a different gateway stays a later option
R14	Last in the ground	Probes and electrodes last 2 growing seasons (about 12 months buried) without replacement	Materials review; later soak test	At risk: hobby probes corrode quickly unless sealed

Summary: 8 met, 5 at risk (R1, R5, R9, R10, R14), 1 not verifiable at TRL 3 (R8), none not met.

■ Assumptions

- Soil water content, not soil water potential, is the quantity measured. Converting it to plant-available water needs field capacity and wilting point for the soil, set during calibration (FAO-56 method).
- EC is measured as bulk soil EC. Pore-water or saturated-paste EC, which crop salinity thresholds use, differ and need a moisture-dependent conversion.
- LoRaWAN uplink payload of 11 bytes (two moisture counts, EC, temperature, cell voltage, status), 24 bytes on air with LoRaWAN overhead.
- SF10 at 125 kHz is the design case because it is the slowest uplink data rate allowed in the US915 plan and a reasonable margin case in EU868.
- The Things Network fair use policy (30 s of uplink airtime per node per day) applies when the community network is used.
- Worst-month solar resource of 3 peak sun hours on the panel before shading and dust losses.